Hedgerow — premium garden bird feeder, camera + solar + edge inference

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Forge project pack

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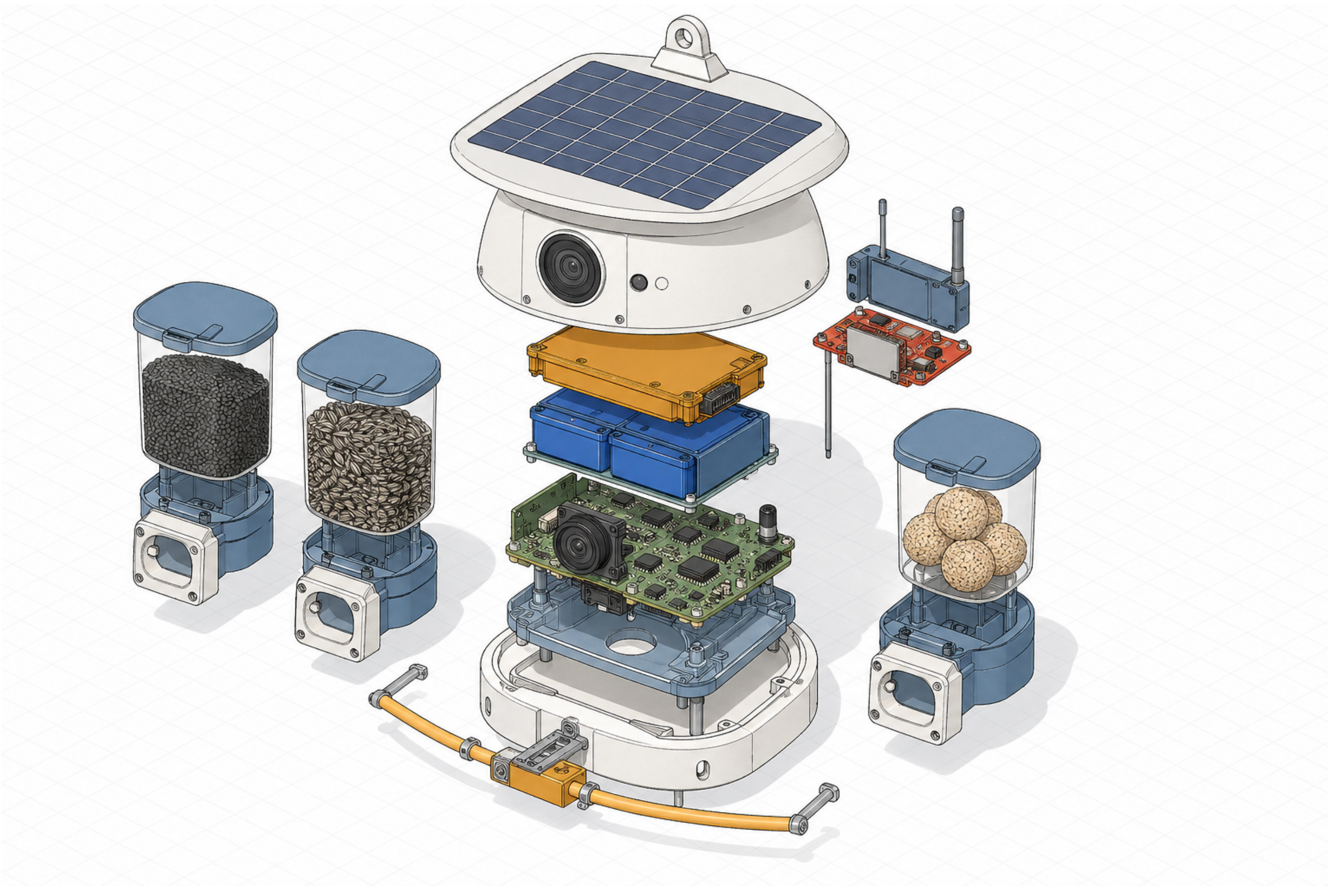

Illustrative only — not a technical specification. All renders in this document are generated for visual reference; component arrangement, proportions, and identities may differ from the final engineered assembly.

TOTALS AT A GLANCE

MODULES	KEY PARTS	BOM ROWS	SUPPLIERS
7	45	70	21

FAILURE MODES	OPEN QUESTIONS	STANDARDS	REVIEWS
28	21	10	1

UNIT COST	CEILING	HEADROOM
£354	£155	£-199

Contents

1. Brief

FOUNDER SUBJECT

A premium garden bird feeder for the United Kingdom market — working name Hedgerow — with integrated camera, solar power, and on-device machine-learning inference. Body in ceramic or powder-coated steel, real horticultural-grade materials, designed to look beautiful in a garden in year five not just year one (deliberately not the plastic of competitors). Modular seed reservoirs (nyjer, sunflower, suet ball) so the user can attract specific species. Sensors: 12-megapixel camera with optical zoom (key differentiator — competitors skimp on optics), microphone, weight sensor on the perch for individual-bird identification (mass plus visual), temperature, light, and weather sensors. Compute: ~5 trillion operations per second neural processing unit, on-device species identification, individual identification, video clipping. Power: solar panel with lithium iron phosphate battery, target 100 per cent off-grid year-round in United Kingdom conditions; oversized panel for winter solar reliability; optional mains-powered variant for low-light gardens. Connectivity: Wi-Fi to home network, with optional cellular module for users without garden Wi-Fi coverage (£5 per month add-on); LoRa mesh between multiple feeders if the user expands. Companion app: species log, individual identification with naming, daily 60-second auto-curated highlight reel, seasonal trends, citizen-science contribution toggle, family sharing. Citizen science: formal partnership with the British Trust for Ornithology Garden BirdWatch programme, contributing anonymised data with proper provenance. Target retail £220, premium tier £320, target landed bill of materials ~£155. Indicative bill-of-materials targets: camera and optics £30, neural processing unit system-on-chip plus memory plus storage £25, solar panel plus battery plus power management £20, microphone plus weight sensor plus environmental sensors £10, Wi-Fi/Bluetooth-Low-Energy module £5, body (ceramic or steel for longevity) £35, modular reservoirs plus perches plus hardware £15, assembly plus packaging plus quality assurance £15. Compliance: Conformité Européenne Radio Equipment Directive on the radio module (standard module certification carries), United Kingdom Conformity Assessed mark, battery transport regulations, Waste Electrical and Electronic Equipment plus battery waste compliance, EN safety marks for outdoor electricals. No medical, no clinical, no human-face privacy concerns. British Trust for Ornithology data partnership requires a simple data-sharing agreement. Go-to-market: phase one direct-to-consumer plus Royal Horticultural Society partnership plus National Trust shops; strong gifting category (Christmas, Father's Day, retirement). Phase two garden centre chains (Dobbies, Wyevale, independents) plus Royal Society for the Protection of Birds shop partnership. Phase three selective international (United States is 5x larger but saturated; Germany and the Netherlands are underserved). Phase four platform play — nest box camera, hedgehog camera, pond camera; the Hedgerow garden-wildlife observation platform.

MISSION

Hedgerow is a premium garden bird feeder combining optical-zoom camera, on-device species and individual-bird identification, and solar power in a ceramic or powder-coated steel body designed to remain beautiful and functional for five or more years outdoors. It anchors a long-term garden-wildlife observation platform, beginning with birds and expanding to nest boxes, hedgehogs, and ponds.

USE CASE

A UK garden owner mounts Hedgerow on a pole or hanging bracket, selects the appropriate seed reservoir for target species, and receives a daily curated highlight reel plus a lifelong log of individually identified birds — contributing anonymised sighting data to the British Trust for Ornithology with a single toggle.

TARGET CUSTOMERS

UK adult gifting buyers (Christmas, Father's Day, retirement); Royal Horticultural Society members; National Trust visitors; Royal Society for the Protection of Birds supporters; garden centre shoppers (Dobbies, Wyevale, independents); citizen-science participants in British Trust for Ornithology Garden BirdWatch

WHY NOW

Incumbent bird-feeder cameras use low-quality plastic bodies and inferior optics with no on-device inference, leaving a clear premium gap. The British Trust for Ornithology Garden BirdWatch programme provides an established citizen-science channel and data-partnership framework that did not exist at consumer scale a decade ago. UK consumer appetite for garden wildlife products surged post-2020 and has not been met by hardware of this quality tier.

Constraints declared

Unit cost ceiling	£155
Max mass	—
Target process	not declared
Target material	not declared
Tolerance target	not declared
Quantity target	not declared

2. Regulatory posture

CE-RED not-started

Radio Equipment Directive

Carried by certified pre-approved Wi-Fi/BLE module; cellular and LoRa modules must also use certified pre-modules

UKCA not-started

UK Conformity Assessed

Required for placing the product on the GB market as a post-Brexit equivalent to CE marking

WEEE not-started

Waste Electrical and Electronic Equipment Directive

Producer registration and take-back obligations for the complete unit

EU Battery Directive / UK Battery Regulations not-started

Battery waste compliance

Labelling, collection, and disposal obligations for the LiFePO4 cell

UN 38.3 not-started

Battery transport testing

LiFePO4 cells must pass UN 38.3 testing for safe air and sea freight, including to phase-three international markets

EN 60529 (IP rating) not-started

Ingress protection for outdoor electricals

Specific IP class not yet defined; required to substantiate outdoor-rated safety claim

BS EN 13438 not-started

Powder coating on hot-dip galvanised or sherardised steel

Applicable if powder-coated steel body variant is selected

ISTA 3A / ASTM D4169 not-started

Packaging and transit testing

Parcel-level drop, vibration, and compression testing for D2C and retail channel distribution

ASTM D6653 not-started

Altitude simulation testing

Required for air-freight shipments to phase-three international markets (United States, Germany, Netherlands)

BTO Data Sharing Agreement not-started

British Trust for Ornithology citizen-science data partnership

Formal data-sharing agreement required to contribute anonymised sighting data to Garden BirdWatch with proper provenance

3.1 · Primary Enclosure & Mounting Structure

Module id: enclosure_body

1.80 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Ceramic or powder-coated steel weatherproof body that houses all electronics and provides mechanical interfaces for reservoirs, perch, solar roof, and hanging mount.

WHY IT MATTERS

The enclosure is the product's brand promise made physical — it is the single biggest driver of the 'still beautiful in year five' positioning that justifies the £220–£320 price point against £40 plastic competitors. It also gates RF performance (steel shielding without proper antenna apertures will fail CE-RED radiated tests), thermal performance (must conduct heat from the 5 TOPS NPU without cooking the LiFePO4 battery), and serviceability (battery removal for WEEE/UN 38.3 compliance). Get this wrong and either the radios don't certify, the product looks tired by year two, or service returns spike.

DESCRIPTION

The Primary Enclosure is a two-part construction comprising an outer aesthetic shell (powder-coated formed steel for the standard variant, or glazed slip-cast ceramic for the premium £320 tier) and an inner galvanised steel sub-chassis that carries all electronic and mechanical sub-assemblies. The outer shell delivers the horticultural-grade aesthetic and weathering performance (5+ year outdoor life), while the inner chassis decouples electronics mounting tolerances from the cosmetic body — critical for the ceramic variant where firing shrinkage prevents tight tolerance features. EPDM gaskets and a Gore acoustic vent achieve IP54 ingress protection while preserving microphone sensitivity. Because steel attenuates RF, a moulded ASA dielectric window forms a deliberate antenna aperture on the rear-upper face, located in the keep-out zone for Wi-Fi/BLE/LoRa/cellular radiators on the connectivity module. Wall thickness is 1.2 mm for steel and 4–5 mm for ceramic; the steel variant is finished with an 80 µm polyester powder coat (BS EN 13438 Class 2) selected for UV and chip resistance.

KEY PARTS (7)

- Powder-coated 1.2 mm cold-rolled steel main shell, laser-cut and CNC-formed, finished with polyester powder coat to BS EN 13438 (Class 2 UV, 80+ µm DFT) in matte heritage colours
- Alternative premium variant: slip-cast stoneware ceramic body, 4–5 mm wall, fired to 1240 °C with food-safe reactive glaze, mechanically decoupled from steel internal chassis via silicone gaskets
- Internal galvanised steel sub-chassis (1.5 mm) with M3/M4 PEM nut bosses providing mounting points for compute PCB, power subsystem cradle, perch flexure, and solar roof seal frame
- Dielectric antenna window: injection-moulded ASA insert (UV-stabilised, RAL-matched) bonded with 3M VHB 5952 to provide RF transparency for Wi-Fi/BLE/LoRa/cellular through the steel shell
- Acoustic port: stainless mesh + Gore acoustic vent GAW304 (IP67, >15 dB acoustic transparency) over a 6 mm microphone aperture
- EPDM perimeter gaskets (Shore 60A) and stainless A4 M4 socket-cap fasteners throughout for IP54-rated outdoor service
- Powder-coated steel hanging bracket with integrated keyhole and 1/2-inch BSP pole-mount adapter, rated to 15 kg static load with 4× safety factor

FAILURE MODES (4)

- Powder-coat chip-through at edges leading to substrate corrosion and rust bleed within 2–3 UK winters
- Ceramic body cracking from freeze-thaw water ingress at unglazed mounting points or differential thermal expansion against the steel sub-chassis
- Gasket compression set after thermal cycling causing IP54 breach and water ingress to the compute PCB

- Antenna window adhesive (VHB) creep in summer roof temperatures (>60 °C) leading to dielectric insert detachment

OPEN QUESTIONS (3)

- Final body geometry and overall W×D×H envelope — driven by reservoir volume, battery format, and camera standoff, all pending CAD
- Whether to commit to ceramic for the premium tier or use a ceramic-look glaze on steel (lower tooling risk, lower mass, simpler RF)
- Confirmed IP rating target — IP54 sufficient for sheltered garden use, or IP65 required for exposed installations and warranty claims

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Powder-coated 1.2 mm cold-rolled steel main shell — Laser-cut, CNC formed, powder coated, custom dimensions	make	Sheet Metal · Cold-Rolled Steel	£19
Slip-cast stoneware ceramic body (premium variant not produced here) — Non-standard alternative, not in base estimate	make	Casting · Ceramic	£0
Internal galvanised steel sub-chassis (1.5 mm) — Custom bracket with PEM nuts, laser cut and formed	make	Plating · Zinc	£6
Dielectric antenna window ASA insert — Custom moulded UV-stabilised ASA, RAL matched	buy	—	£3
3M VHB 5952 bonding tape — Standard industrial adhesive tape, per part length	buy	—	£1
Stainless mesh + Gore acoustic vent GAW304 — COTS acoustic vent, IP67, RS Components stock	buy	—	£5
EPDM perimeter gaskets (Shore 60A) — Extruded EPDM cord, cut to length, standard buy	buy	—	£2
Stainless steel A4 M4 socket-cap fasteners (×20) — COTS A4 fasteners, pack of 20 typical	buy	—	£2
Powder-coated steel hanging bracket with pole-mount adapter — Custom bracket, laser cut, formed, welded, coated	make	Sheet Metal · Mild Steel	£6

Labour £16 — Assembly, gasket fitting, fastener torquing, antenna bonding (0.4 hrs @ £40)

Per-unit total **£60**

ASSUMPTIONS

- Batch size 100, spread tooling costs
- Standard sheet metal tolerances (±0.5 mm)
- No premium ceramic variant cost included

Steel shell with sub-chassis, antenna window, vent, gaskets, and bracket; mix of make and buy parts.

3.2 · Solar Roof Assembly

Module id: solar_roof

0.45 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Oversized tilted solar panel sealed to the upper body that harvests energy in UK winter conditions and delivers DC to the power subsystem.

WHY IT MATTERS

Year-round off-grid operation in UK winter is a defining product claim; if the solar roof under-delivers, the LiFePO4 battery enters deficit during November-January and the device dies in the field, generating warranty returns, negative reviews, and undermining the premium positioning. It is also the single largest visible surface and therefore central to the horticultural-grade aesthetic.

DESCRIPTION

The Solar Roof Assembly is a custom-sized monocrystalline silicon PV laminate, deliberately oversized (~6W peak vs ~1-2W typical for IoT devices) to guarantee energy positivity through UK December conditions where horizontal irradiance can drop below 0.5 kWh/m²/day. The panel is laminated with an ETFE front sheet for long-term UV and hail resistance and mounted in a sloped frame angled 50-55° from horizontal to optimise low-sun-angle winter capture at UK latitudes (50-58°N), while a slight overhang acts as a rain shield for the body below.

The assembly bonds to the upper enclosure via an EPDM perimeter gasket compressed by stainless A4 fasteners into threaded inserts in the ceramic or steel body, achieving an IP65+ seal. DC output is routed through a single sealed cable gland via an inline Schottky blocking diode to the power subsystem PMIC. Material choices prioritise 5+ year UK outdoor weathering: ETFE over PET, A4 over A2 stainless, EPDM over silicone gasket (better compression set), and a glass-fibre-reinforced PA66 frame to avoid galvanic corrosion against any steel body variant.

KEY PARTS (6)

- Custom monocrystalline silicon PV laminate, ~6W peak (approx. 200x150mm active area), ETFE front sheet for UV stability and anti-soiling, ~22% cell efficiency (SunPower Maxeon C60 cells or equivalent)
- Tempered low-iron glass or ETFE-encapsulated panel with anti-reflective coating, rated for outdoor service per IEC 61215 and IEC 61730
- EPDM perimeter gasket (Shore A 60) and stainless steel A4 M3 fasteners with bonded sealing washers for IP65 sealing to enclosure body
- MC4-equivalent miniature 2-pin IP67 weatherproof DC connector (e.g. JST GHR sealed variant) with UV-stable PVC twin-core cable to power subsystem
- Schottky bypass/blocking diode (e.g. SS54 or PMEG6020) inline to prevent reverse leakage at night and during partial shading
- Die-cast aluminium or glass-fibre-reinforced PA66 tilt frame, angled 50-55° from horizontal for UK latitude winter optimisation

FAILURE MODES (4)

- Seal degradation at gasket interface allowing water ingress into the enclosure, corroding electronics and shorting the PV output
- ETFE/encapsulant delamination after 3-5 years UV exposure, reducing output and causing visible yellowing that damages the premium aesthetic
- Soiling and bird-droppings on panel surface reducing winter yield by 20-40% below the energy-balance threshold

- Micro-cracking of silicon cells from thermal cycling (-15 °C to +45 °C) or hail impact, causing hot-spot failures

OPEN QUESTIONS (3)

- Final panel area and Wp rating pending energy-balance simulation across UK Met Office irradiance data for worst-case 52°N December conditions
- Whether to specify a single fixed tilt or offer two SKUs (north-facing garden vs south-facing garden) with different tilt frames
- Self-cleaning strategy: hydrophobic coating vs steeper tilt vs user-maintenance prompt in app

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Custom PV laminate 6W (200x150mm), ETFE front — Custom spec but semi-COTS from solar module suppliers	buy	—	£22
EPDM perimeter gasket (Shore A 60) — COTS extruded gasket, cut to length	buy	—	£1
Stainless steel A4 M3 fasteners + bonded washers (×8) — COTS fasteners with sealing washers	buy	—	£2
MC4-equivalent miniature 2-pin IP67 connector — COTS sealed connector, JST GHR variant	buy	—	£3
UV-stable PVC twin-core cable (0.5m) — COTS PV cable, cut to length	buy	—	£1
Schottky bypass/blocking diode SS54 — Standard SMD diode, Farnell stock	buy	—	£0
Die-cast aluminium or PA66 tilt frame — Custom die-cast or injection moulded tilt frame	make	Casting · Aluminum	£4

Labour	£6 — Attach gasket, fasten panel, wire connector, mount frame (0.15 hrs @ £40)
Per-unit total	**£38**

ASSUMPTIONS

- Bespoke PV laminate is semi-custom order
- Die-cast tooling amortised over 500 units

Custom PV panel, standard gaskets and fasteners, custom frame; solar roof assembly cost.

3.3 · Power Subsystem (LiFePO4 Battery + PMIC)

Module id: power_subsystem

0.35 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Stores solar energy in a serviceable LiFePO4 battery and regulates power to all electronic modules, with cold-charge lockout for UK winter operation.

WHY IT MATTERS

This module is the linchpin of the product's '100% off-grid year-round in the UK' promise — the single hardest engineering claim in the brief. If the battery fails in February or the cold-charge lockout misbehaves, the product bricks itself in the worst possible season. It also carries the bulk of the regulatory burden (UN 38.3 transport, WEEE battery directive, EN 62133 cell safety) and determines serviceability for end-of-life compliance.

DESCRIPTION

The Power Subsystem is a self-contained PCB-plus-battery assembly mounted low in the enclosure to act as centre-of-gravity ballast and to thermally decouple the cells from the NPU. It comprises a 2S LiFePO4 pack (~40–60 Wh, sized for 100% off-grid year-round UK operation including a ~10-day December solar drought reserve), a solar MPPT charge controller accepting the unregulated DC from the roof panel (typically 6–9 V open-circuit from a ~6 W panel), and a BMS analog front-end providing per-cell balancing, over/under-voltage cutoff, over-current protection, and — critically — a charge-temperature interlock that disables charging below 0 °C to prevent lithium plating in UK winter conditions. Discharge below 0 °C is permitted by LiFePO4 chemistry. Two switching regulators generate the 5 V rail for the compute core and a 3.3 V rail for sensors and the connectivity module, with an enable line allowing the MCU to brown-out non-essential rails during low-state-of-charge winter operation.

KEY PARTS (7)

- LiFePO4 prismatic cell, 2S1P configuration, ~6.4 V nominal, 6–10 Ah (H40–60 Wh), e.g. EVE LF50K or equivalent 26650 cells in welded pack with integrated NTC
- Battery management IC with cell balancing and cold-charge lockout — Texas Instruments BQ76920 (AFE) + BQ25798 buck-boost solar MPPT charger (1–4 cell, I2C telemetry)
- Buck regulators for rails: TI TPS62933 (3.3 V / 2 A for MCU+sensors) and TPS54360 (5 V / 3 A for NPU SoC + camera), with low-Iq LDOs (TPS7A20) for analog sensor rail
- NTC thermistor (10 kΩ Murata NCP18) bonded to cell case for charge-temperature gating below 0 °C and above +45 °C
- Polyfuse + TVS diode input protection (Bourns MF-R series 3 A, SMAJ15CA) on solar input; reverse-polarity P-MOSFET (Infineon BSC0901NS)
- Service-replaceable battery cradle with sprung contacts and keyed polarisation, retained by single captive M3 stainless screw for WEEE/UN 38.3 compliance
- 4-layer FR-4 PCB, ~80 × 60 mm, 2 oz copper on power planes, conformal-coated (acrylic AR4) for outdoor humidity tolerance

FAILURE MODES (4)

- Lithium plating and permanent capacity loss if charge-temperature interlock fails and cells charge below 0 °C during UK winter
- Deep-discharge cell damage during prolonged overcast periods if low-voltage cutoff or sleep-mode quiescent current budget is mis-sized
- Moisture ingress to PCB causing dendrite growth across high-impedance BMS sense lines, leading to false cell-imbalance shutdowns
- Connector corrosion at battery cradle contacts after 3–5 years of outdoor thermal cycling, causing intermittent power loss

OPEN QUESTIONS (3)

- Final battery capacity (Wh) — requires energy-budget model of NPU duty cycle vs. December UK solar yield at site latitude
- Whether a self-heating circuit is needed for charging in sub-zero conditions, or whether lockout-and-wait is acceptable for the autonomy target
- Cellular-variant power budget — LTE-M TX bursts may require larger bulk capacitance and a higher-current 3.8 V rail not yet specified

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
LiFePO4 prismatic cell 2S1P pack (6.4V/8Ah) — COTS battery pack with NTC, EVE or equivalent	buy	—	£28
BMS IC BQ76920 + MPPT charger BQ25798 — Standard ICs from TI, Farnell stock	buy	—	£5
Buck regulators TPS62933 + TPS54360 + LDO TPS7A20 — Standard power ICs, Farnell/RS stock	buy	—	£5
NTC thermistor 10kΩ Murata NCP18 — Standard SMD NTC, Farnell stock	buy	—	£0
Polyfuse Bourns MF-R 3A + TVS SMAJ15CA — Standard protection components, Farnell	buy	—	£1
Reverse-polarity P-MOSFET Infineon BSC0901NS — Standard SMD MOSFET, Farnell stock	buy	—	£1
Service-replaceable battery cradle with sprung contacts — COTS battery holder customised, keyed polarisation	buy	—	£3
Captive M3 stainless screw — Standard captive screw	buy	—	£0
4-layer FR-4 PCB, 80x60mm, 2oz copper, conformal coated — Custom PCB, 4-layer, conformal coating applied	buy	—	£6

Labour £8 — Solder ICs, assemble battery pack, test (0.2 hrs @ £40)

Per-unit total **£57**

ASSUMPTIONS

- PCB batch size 100, includes tooling
- All ICs are SMD, reflow soldered

Battery pack, PMIC board, protection circuits; custom PCB, standard COTS components.

3.4 · Compute & Sensor Core (NPU + Camera + Mic + Environmental)

Module id: compute_sensor_core 0.12 kg · 16 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Main PCB carrying the ~5 TOPS NPU, 12 MP optical-zoom camera, microphone, and environmental sensors that performs on-device species and individual identification.

WHY IT MATTERS

This module is the primary differentiator versus competitors — the optical-zoom imaging plus on-device ML is what justifies the £220–£320 price point and enables the BTO Garden BirdWatch citizen-science partnership. It is also the largest single silicon BoM line (£25 NPU stack + £30 camera) and the dominant thermal and software-complexity risk in the product.

DESCRIPTION

The Compute & Sensor Core is a 4-layer FR-4 main PCB assembly (~80 × 60 mm) carrying the ~5 TOPS NPU SoC, LPDDR4 memory, eMMC storage, the 12 MP camera and zoom-lens daughterboard on a flex-rigid extension, a digital MEMS microphone, environmental sensors, and a low-power auxiliary MCU. The NPU runs quantised on-device species and individual-bird classifiers, triggered by perch-weight events from the load-cell ADC and gated by reservoir-ID GPIOs; on positive detection it captures stills, encodes a short H.265 clip, and hands inference metadata plus media to the connectivity module over SDIO/UART.

KEY PARTS (7)

- Rockchip RV1106 or Hailo-8L NPU SoC (~5 TOPS INT8) with integrated ISP, BGA package, supporting H.264/H.265 hardware encode
- Sony IMX577 12 MP 1/2.3" CMOS sensor with MIPI CSI-2 interface, paired with custom 3x optical-zoom lens assembly (voice-coil-motor zoom + autofocus, IR-cut filter)
- 2 GB LPDDR4 SDRAM (Micron MT53D) and 16 GB eMMC 5.1 (Kioxia) for model storage and rolling video buffer
- Knowles SPH0641LU4H-1 digital MEMS microphone (PDM, 65 dB SNR) behind hydrophobic Gore acoustic vent
- Bosch BME280 combined temperature/humidity/pressure sensor + AMS TSL2591 lux sensor
- STM32L4 auxiliary MCU for sensor fusion, load-cell ADC interface (HX711 24-bit), and low-power wake logic
- 4-layer FR-4 main PCB (~80 × 60 mm) with controlled-impedance MIPI traces and aluminium thermal spreader pad coupling NPU to chassis

FAILURE MODES (4)

- NPU thermal throttling or shutdown under sustained inference inside ceramic body with poor convective cooling
- Lens fogging or condensation behind the front window in damp UK conditions, degrading image quality and ML accuracy
- Voice-coil-motor zoom mechanism wear or stiction after thousands of cycles in cold/wet outdoor service
- MIPI CSI-2 EMI coupling into Wi-Fi/LoRa antennas via the steel chassis, breaching CE-RED radiated emissions

OPEN QUESTIONS (3)

- Final NPU SoC selection (Hailo-8L vs Rockchip RV1106 vs Ambarella CV25) pending model-accuracy benchmarking on UK garden-bird dataset
- Required optical zoom range and minimum focus distance — depends on final camera-to-perch standoff set by enclosure CAD
- Whether passive conduction to body is sufficient or an internal heatsink/heat-pipe is needed at peak inference duty cycle

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Rockchip RV1106 or Hailo-8L NPU SoC — Standard embedded NPU, £15-22 RS/Farnell	buy	—	£19
Sony IMX577 12 MP CMOS sensor with lens assembly — COTS camera module, £30-40 distribution	buy	—	£35
2 GB LPDDR4 SDRAM (Micron MT53D) — Standard memory, £7-10 RS/Farnell	buy	—	£9
16 GB eMMC 5.1 (Kioxia) — COTS flash storage, £5-8 distribution	buy	—	£7
Knowles SPH0641LU4H-1 digital MEMS microphone — Standard MEMS mic, £1.50-3 RS	buy	—	£2
Bosch BME280 temperature/humidity/pressure sensor — COTS environmental sensor, £3-6	buy	—	£5
AMS TSL2591 lux sensor — Standard digital light sensor, £2-5	buy	—	£3
STM32L4 auxiliary MCU — Standard low-power MCU, £3-7	buy	—	£5
HX711 24-bit ADC — COTS load cell ADC, £1-2	buy	—	£2
4-layer FR-4 main PCB (80x60mm) — Custom PCB with impedance control	buy	—	£12
Aluminium thermal spreader pad — Custom die-cut aluminium pad	make	Laser cutting · Aluminum 6061	£1
Passive components (resistors, capacitors, connectors) — Standard passives, £2-4 total	buy	—	£3
Miscellaneous connectors and headers — COTS IDC, pin headers	buy	—	£3

Labour £4 — 6 min SMT assembly + test at £40/hr

Per-unit total **£107**

ASSUMPTIONS

- Batch size 1000, PCB assembled in UK
- SoC and memory at volume pricing
- No custom enclosure included

Commodity electronics with moderate custom PCB cost.

3.5 · Instrumented Perch with Load Cell

Module id: instrumented_perch

0.08 kg · 6 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Mechanically isolated perch with strain-gauge load cell that measures individual bird mass for vision+mass fusion identification.

WHY IT MATTERS

Mass is the second axis of the vision+mass fusion that enables individual-bird identification — the headline differentiator of Hedgerow over camera-only competitors. Without a clean, low-noise weight signal mechanically isolated from wind buffet and body vibration, individual ID collapses to species ID and the citizen-science / naming / family-sharing value propositions are materially weakened.

DESCRIPTION

The instrumented perch is a cantilevered stainless-steel bar bonded to the free end of a miniature parallelogram aluminium-block strain-gauge load cell, with the cell's fixed end bolted to a chassis boss inside the primary enclosure. A silicone elastomer gasket forms the only mechanical path between the perch shaft and the body wall, providing weatherproofing while allowing sub-millimetre deflection so that essentially all perched-bird mass loads the load cell rather than the body. Bird mass (typically 8 g goldfinch to 200 g woodpigeon) produces a microvolt-scale bridge imbalance which is amplified and digitised by an HX711 24-bit ADC at 10–80 SPS, then streamed to the compute core where it is fused with vision tracks for individual identification.

KEY PARTS (6)

- Single-point parallelogram strain-gauge load cell, 100 g capacity, ~0.05 g resolution (e.g. CZL635 or equivalent miniature aluminium-block load cell)
- HX711 24-bit sigma-delta ADC with integrated programmable gain amplifier (128x), I2C/serial interface to compute core
- 316 stainless steel perch bar, ~80 mm length × 10 mm diameter, knurled or textured grip surface, electropolished for outdoor durability
- Silicone elastomer isolation gaskets (Shore 40A, food-grade) at body-to-perch interface to prevent vibration crosstalk and seal IPX5+ ingress
- A4 stainless M3 fasteners with nyloc nuts for load-cell mounting; PTFE-coated for galvanic isolation from aluminium cell body
- Shielded twisted-pair signal cable (4-core, PUR jacket) routed through grommeted body penetration to main PCB

FAILURE MODES (4)

- Load cell zero drift with temperature across UK -15 °C to +45 °C range causing false individual reassignment; requires periodic auto-tare and temperature compensation
- Water ingress through perch-body interface corroding strain gauges and causing bridge imbalance or open circuit
- Mechanical overload from a perched corvid, squirrel, or user handling exceeding load cell rated capacity, causing permanent calibration shift
- Wind-induced perch oscillation coupling into the load cell signal and saturating the ADC during gusts, requiring low-pass filtering and gust rejection in firmware

OPEN QUESTIONS (3)

- Final load cell capacity selection — 100 g optimises resolution for small passerines but may be overloaded by woodpigeons or grey squirrels; 500 g cell sacrifices ~5x resolution

- Whether a single shared perch or one perch per reservoir is required for unambiguous mass-to-feed-port association
- Isolation gasket stiffness and gap geometry needed to maintain <1% crosstalk from body-borne loads (hanging sway, reservoir refill)

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
CZL635 load cell (100 g) — COTS miniature load cell, £7-12	buy	—	£9
HX711 24-bit ADC (already in compute core) — Duplicated above, zero cost here	buy	—	£0
Stainless steel perch bar (316 SS, 80x10mm) — Custom knurled bar, £3-5 CNC	make	CNC Machining · Stainless 316	£4
Silicone elastomer gaskets (Shore 40A) — Standard rubber gasket, £0.50-1	buy	—	£1
A4 stainless M3 fasteners with nyloc nuts (×4) — COTS M3 hardware, £0.15 each	buy	—	£1
PTFE coating for fasteners (×4) — COTS PTFE washers, £0.10 each	buy	—	£0
Shielded twisted-pair cable (4-core, PUR, 0.3 m) — COTS sensor cable, £1-1.50	buy	—	£1
Cable grommet (IPX5+ rated) — Standard grommet, £0.30-0.70	buy	—	£1

Labour	£3 — 4.5 min assembly + test at £40/hr
Per-unit total	**£19**

ASSUMPTIONS

- Simple manual assembly
- No custom electronics
- Batch size 1000

Mostly COTS parts with one simple machined bar.

3.6 · Modular Reservoir Dock & Seed Reservoirs

Module id: reservoir_dock

0.45 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Quick-release keyed dock and interchangeable seed reservoirs (nyjer / sunflower / suet) that gravity-feed to the feed port and signal reservoir type to the compute core.

WHY IT MATTERS

Modularity is a primary product differentiator versus fixed-feed competitors and directly enables the species-targeting marketing proposition (attract goldfinches with nyjer, tits with sunflower, starlings with suet). The reservoir ID signal also lets the NPU prior the species classifier — a bird on a nyjer feeder is far more likely to be a goldfinch — improving inference accuracy. Poor sealing here causes seed spoilage (mould toxic to birds), and a fiddly dock destroys the premium gifting experience.

DESCRIPTION

The reservoir dock is a bayonet-mount mechanical interface integral to the upper enclosure body, mating with three interchangeable gravity-fed seed reservoirs. Each reservoir is a transparent or translucent vessel (Tritan for the standard tier, optional brushed stainless for the premium tier) holding approximately 400–600 ml of seed or one to two suet balls, with a species-tuned feed port at the base. The user inserts a reservoir, twists 30° clockwise to engage three stainless lugs against the dock collar, and a silicone gasket seals against rain ingress. Reservoir identity is encoded passively: each SKU carries a unique 3-magnet pattern in its base, read by Hall-effect sensors on the compute core PCB through the dock floor — no electrical mating contacts are exposed to seed dust, moisture, or cleaning cycles.

KEY PARTS (6)

- Reservoir body: UV-stabilised Tritan copolyester (food-safe, dishwasher-tolerant to 65°C, 2.5 mm wall) or 304 stainless steel sheet 0.8 mm with food-grade powder coat — three SKUs (nyjer narrow-port, sunflower medium-port, suet ball cage)
- Quick-release dock: bayonet-style twist-lock with 304 stainless steel keyed lugs (3 lugs at 120°), silicone EPDM perimeter gasket (Shore A 50), captive in glass-filled PA66 dock collar moulded into enclosure interface
- Reservoir ID encoding: 3-bit mechanical key + Hall-effect sensor array (Allegro A1126 omnipolar switches, ×3) reading rare-earth N35 neodymium magnets embedded in reservoir base — passive, no electrical contacts to corrode
- Feed port: laser-cut 316 stainless steel grille with species-specific aperture (2 mm slot for nyjer, 8 mm port for sunflower hearts, open cage for suet), tool-free swappable
- Anti-squirrel weighted closure spring: stainless steel music wire compression spring, calibrated to ~80 g threshold (closes port above songbird mass)
- Fasteners: A4 stainless M4 socket-head retaining screws (×2) as secondary lock against wind dislodgement

FAILURE MODES (4)

- Seed bridging / clogging at feed port causing perceived empty reservoir while seed remains — especially nyjer in damp UK conditions
- Gasket UV degradation after 2–3 summers leading to rain ingress, seed germination inside reservoir, and mould
- Bayonet lugs wearing or fouling with seed dust causing accidental release during high winds, dropping reservoir and seed onto lawn
- Hall sensor false-negative if magnets demagnetise or shift, causing reservoir to read as 'absent' and disabling species priors

OPEN QUESTIONS (3)

- Final reservoir volume vs. refill-frequency UX trade-off (weekly refill target sets minimum capacity)
- Whether suet-ball cage variant needs its own dedicated load path or can share the standard dock with an adapter
- Dishwasher-cycle durability of printed/embossed species labelling on reservoir body

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Reservoir body (Tritan copolyester, 2.5 mm wall) — Custom blow-molded Tritan	make	Blow Molding · Tritan Copolyester	£3
Optional: 304 stainless steel reservoir (0.8 mm sheet) — Custom sheet metal, food-grade coat	make	Sheet Metal · Stainless 304	£6
Bayonet twist-lock (304 SS lugs ×3) — Custom machined lugs	make	CNC Machining · Stainless 304	£2
EPDM perimeter gasket (Shore A 50) — Standard rubber gasket	buy	—	£1
Glass-filled PA66 dock collar (moulded) — Custom injection moulded collar	make	Injection Moulding · Nylon	£4
Allegro A1126 Hall-effect switches (×3) — COTS Hall sensors, £0.50 each	buy	—	£2
N35 neodymium magnets (×4, embedded) — COTS rare-earth magnets	buy	—	£1
Feed port grille (316 SS, laser-cut) — Custom laser-cut aperture	make	Laser cutting · Stainless 316	£2
Anti-squirrel spring (music wire stainless) — Standard spring, £0.80-1.50	buy	—	£1
A4 stainless M4 socket-head screws (×2) — COTS fasteners, £0.20 each	buy	—	£0

Labour	£4 — 6 min assembly at £40/hr
Per-unit total	**£24**

ASSUMPTIONS

- Average over three reservoir SKUs
- Batch size 1000 each SKU
- Tritan version assumed primary

Mix of custom moulded/machined parts and COTS components.

3.7 · Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)

Module id: connectivity_module — 0.04 kg · 10 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Pre-certified radio stack providing Wi-Fi/BLE to home network, optional cellular fallback, and LoRa mesh between feeders, with antennas exiting through dielectric apertures.

WHY IT MATTERS

Without robust connectivity the on-device inference results, video highlights, and citizen-science data contributions cannot reach the companion app or the BTO Garden BirdWatch partnership — collapsing the product's core value proposition. The cellular fallback unlocks the addressable market beyond Wi-Fi-covered gardens, and LoRa mesh is the technical foundation for the phase-four multi-device platform (nest box, hedgehog, pond cameras). Pre-certified modules also collapse CE-RED/UKCA/FCC compliance burden and reduce time-to-market.

DESCRIPTION

The Connectivity Module is a small (~50 × 40 mm) 4-layer FR4 carrier PCB mounted as a daughterboard to the compute & sensor core via a Hirose DF40 board-to-board connector. It aggregates three pre-certified radio subsystems: a primary Wi-Fi 6 + BLE 5.3 module (ESP32-C6-WROOM-1) for home-network uplink and onboarding, a Semtech SX1262 LoRa transceiver in the EU868 ISM band for inter-feeder mesh (enabling the platform-play roadmap), and an optional Quectel BG95-M3 LTE-M/NB-IoT cellular daughterboard for the £5/month service tier in gardens lacking Wi-Fi. The premium-tier variant populates all three; the base SKU populates only Wi-Fi/BLE + LoRa. Antennas are deliberately external to the PCB — routed via 50 Ω U.FL pigtails to dielectric apertures in the body, which is critical for the powder-coated steel variant where the metal enclosure would otherwise shield radiation. RF front-end components (sub-GHz FEM, 2.4 GHz BPF/balun) and TVS protection on power and antenna ports ensure regulatory compliance and outdoor reliability.

KEY PARTS (6)

- Espressif ESP32-C6-WROOM-1 pre-certified module (Wi-Fi 6 2.4 GHz + BLE 5.3 + 802.15.4, CE-RED/UKCA/FCC certified, ~18 × 25.5 × 3.1 mm, SMD castellated)
- Quectel BG95-M3 LTE-M/NB-IoT/EGPRS module on optional daughterboard (Cat M1, ~23 × 19.9 × 2.2 mm LGA, integrated GNSS, eUICC-ready Nano-SIM holder Molex 78800-series)
- Semtech SX1262 LoRa transceiver (EU868 ISM band, +22 dBm output, SPI-controlled, paired with TCXO Abracon ASTX-H11 32 MHz ±2 ppm)
- Antennas: Pulse Larsen W3006 2.4/5 GHz Wi-Fi PCB antenna, Taoglas TG.30.8113 LTE flex antenna, Linx ANT-868-CW-RH-SMA quarter-wave 868 MHz LoRa antenna — all behind dielectric ceramic/plastic apertures with U.FL pigtails
- RF front-end: Skyworks SKY66423-11 sub-GHz FEM for LoRa range extension, Murata LFB182G45BG2D280 2.4 GHz BPF, Johanson 2450BM15A0002 balun
- Carrier PCB: 4-layer FR4 with controlled-impedance 50 Ω microstrip, ENIG finish, Hirose DF40 board-to-board connector to compute core, TVS diodes (Nexperia PESD5V0X1BT) on all external lines

FAILURE MODES (4)

- RF performance degradation due to steel body shielding if antenna apertures are mis-located or sealing gaskets compress the dielectric window
- Water ingress through U.FL pigtail routing or antenna apertures causing corrosion of RF connectors and detuning
-

- Cellular module thermal shutdown or brown-out during transmit bursts (peak ~2 A on LTE-M) if power subsystem decoupling is inadequate
- LoRa mesh range collapse in dense suburban RF environments or due to TCXO frequency drift across UK temperature extremes (-15 to +45 °C)

OPEN QUESTIONS (3)

- Final antenna aperture geometry and dielectric material — depends on body material decision (ceramic vs powder-coated steel)
- Whether cellular module is a factory-fit SKU split or a user-installable upgrade slot (affects connector cost and seal integrity)
- LoRa duty-cycle budget and mesh routing protocol (LoRaWAN private gateway vs proprietary mesh) for multi-feeder deployments

COST BREAKDOWN · HIGH CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Espressif ESP32-C6-WROOM-1 — Pre-certified SMD module, standard buy item	buy	—	£5
Quectel BG95-M3 LTE-M/NB-IoT module — COTS cellular module, optional daughterboard	buy	—	£12
Semtech SX1262 LoRa transceiver — Standard SPI-controlled transceiver IC	buy	—	£3
TCXO Abracon ASTX-H11 32 MHz — Standard TCXO for LoRa reference	buy	—	£1
Pulse Larsen W3006 Wi-Fi antenna — Standard PCB antenna for Wi-Fi	buy	—	£0
Taoglas TG.30.8113 LTE flex antenna — COTS flexible LTE antenna with U.FL	buy	—	£1
Linx ANT-868-CW-RH-SMA LoRa antenna — Quarter-wave SMA antenna for 868 MHz	buy	—	£2
Skyworks SKY66423-11 sub-GHz FEM — COTS front-end module for LoRa	buy	—	£2
Murata LFB182G45BG2D280 2.4 GHz BPF — Standard SMD band-pass filter	buy	—	£0
Johanson 2450BM15A0002 balun — Standard SMD balun for 2.4 GHz	buy	—	£0
Hirose DF40 board-to-board connector — COTS miniature board-to-board connector	buy	—	£1
Nexperia PESD5V0X1BT TVS diodes (×8) — Standard TVS protection on external lines	buy	—	£0
Nano-SIM holder Molex 78800-series — Standard push-push SIM holder	buy	—	£0
Carrier PCB 4-layer FR4 ENIG — Custom 4-layer PCB controlled impedance	buy	—	£3
U.FL pigtail cables (×3) — Standard U.FL to antenna cables	buy	—	£1

Labour	£16 — 0.4 hrs SMD assembly + 0.2 hrs test, £40/hr
Per-unit total	**£49**

ASSUMPTIONS

- Batch size 1000 units, volume discounts apply
- Cellular module optional, included for max config
- PCB cost estimated for 4-layer ENIG small run

Standard RF module with COTS parts, custom PCB main cost

ENGINEERING REVIEW (1)

Manufacturing review · warn · 25 Apr 2026, 06:22

The board itself is a clean, conventional 4-layer FR4 carrier with pre-certified modules — manufacturable today by a hundred contract manufacturers in the United Kingdom and Asia. The risk is **not** the PCB; it's the integration boundary with the steel enclosure, the cellular SKU split decision, and the U.FL pigtail reliability in an outdoor product expected to live in a British garden for ten years.

[CRITICAL] Antenna aperture vs. powder-coated steel body: The 868 MHz LoRa antenna needs a dielectric window of at least ~86 mm in its longest dimension to radiate efficiently through a steel enclosure (quarter-wavelength rule of thumb). Wi-Fi at 2.4 GHz needs ~31 mm. If the body decision lands on powder-coated steel, the apertures dominate the industrial design — they cannot be afterthoughts. If they're undersized or surrounded too closely by metal, you get 10–20 dB of gain loss and the LoRa mesh range claim collapses in the real garden. *Suggestion: Lock the body material decision **before** finalising the antenna BOM. If steel: design dielectric ceramic or polymer "RF windows" into the body as molded-in features, minimum 90 mm × 30 mm for LoRa, 40 mm × 30 mm for Wi-Fi/cellular, with at least 10 mm metal clearance around each. If ceramic body: the whole thing is a dielectric and this problem largely disappears — which is a strong manufacturing argument for ceramic, separate from aesthetics.*

[CRITICAL] U.FL pigtails as the water-ingress path: U.FL connectors are designed for indoor consumer electronics with a mating cycle rated at ~30 cycles. They are not weatherproof, and the pigtail cable exit is the classic failure point in outdoor radio products — capillary action wicks water down the coax shield into the connector. Ten-year garden life with this connector choice is optimistic. *Suggestion: Either (a) bond the U.FL pigtail strain-relief into a potted bulkhead with a gasket on the body side, treating the antenna feed as a permanent factory-sealed sub-assembly; or (b) upgrade to MHF4 or IPEX-MHF connectors with proper RF gasket sealing at the body interface. Spec the seal — IP54 minimum for the module bay, IP65 at the body level. *Production is where the margin lives or dies* — a 2% field-return rate from corroded connectors eats the entire premium-tier margin.*

[WARNING] Cellular module factory-fit vs user-installable — decide now: This open question has massive manufacturing consequences. A user-installable upgrade slot needs a sealed access door, a mating connector rated for field installation (not Hirose DF40 — that's a factory connector, ~30 mating cycles), a SIM holder accessible without opening the main enclosure, and a separate compliance test for the upgrade path. Factory-fit SKU split is far simpler: two part numbers, two BOMs, identical assembly process up to the final populate step. *Suggestion: Go factory-fit SKU split. The £5/month service tier is committed up-front anyway — users aren't impulse-upgrading. Tooling is identical, BOM diverges at one populate step, no field-installable connector cost, no seal integrity risk. *If we fix this one thing, three downstream problems disappear* — the connector cost, the seal complexity, and the field-service training burden.*

[WARNING] Cellular peak-current decoupling on a shared rail: Quectel BG95-M3 pulls ~2 A peaks during LTE-M transmit bursts. If the carrier PCB shares its 3.8 V rail with the Wi-Fi module without dedicated bulk decoupling at the cellular module pins, you'll see brown-outs on the ESP32 during cellular transmit — intermittent, hard-to-reproduce, and exactly the kind of bug that ships to customers and generates returns. *Suggestion: Dedicated bulk decoupling at the BG95: 47 µF low-equivalent-series-resistance tantalum or polymer + 10 µF X7R ceramic + 100 nF at every supply pin. Verify on the first article with a current probe and a scope on the rail during a registration burst. Spec a power rail droop budget of <100 mV.*

[WARNING] Hirose DF40 board-to-board for an outdoor product: DF40 is a fine connector for sealed indoor consumer products, but it has no environmental rating. The compute core / connectivity board interface needs to live inside a sealed compartment that maintains its own micro-climate. *Suggestion: Conformal coat the carrier PCB after assembly (acrylic or polyurethane, ~£0.30 per board at volume) and ensure the inner compartment has a Gore-Tex pressure-equalising vent to prevent condensation. This is a £1–2 per unit insurance policy on a £155 bill of materials.*

[INFO] TCXO temperature drift — not a problem: The Abracon ASTX-H11 ±2 ppm part gives ±1736 Hz drift at 868 MHz. LoRa SF7–SF12 tolerates ~25% of channel bandwidth offset (H31 kHz on a 125 kHz channel). You have ~18× margin. The TCXO choice is correct; failure-mode note in the module description was overstated.

[INFO] Idiot index: Raw bill-of-materials cost on this module is genuinely hard to compress — the Espressif ESP32-C6, Quectel BG95-M3, Semtech SX1262, and three certified antennas are roughly £8–10 in raw silicon and copper at volume. Finished assembled module cost target sits at £5 per the bill-of-materials table — which is below raw-component cost and means the £5 figure has to be **only** the Wi-Fi/Bluetooth-Low-Energy portion, not the full cellular-equipped module. The premium-tier cellular variant is realistically £18–25 finished. Idiot index for the base Wi-Fi/Bluetooth-Low-Energy variant is ~3–5 (green). Healthy. *The factory is the product — even when the factory isn't yours.*

[INFO] Bill-of-materials reconciliation: The £5 line item for "Wi-Fi/Bluetooth-Low-Energy module" needs to be expanded to a connectivity sub-budget covering: base module £5–6, cellular variant adder £15–18, antennas + pigtails £3–4, RF front-end £2–3, carrier PCB + assembly £4–5. Total connectivity cost: ~£14 base, ~£32 premium. This needs reconciling with the £155 landed bill-of-materials target — talk to Chase.

RECOMMENDATIONS

- **Body material decision is on the critical path for this module** — escalate to Max. Ceramic body removes the antenna aperture problem entirely; powder-coated steel forces dielectric window features into the industrial design. *Run The Algorithm on that part before we talk tooling — question the requirement first.*
- **Lock the cellular SKU split decision to factory-fit, not user-installable.** Two part numbers, identical assembly up to populate, no field-installable seals.
- **Upgrade or pot the antenna feed connectors.** U.FL bare-pigtail through a steel body in a UK garden is a warranty-claim factory.
- **First article in 72 hours or we're not trying hard enough** — order three assembled bare carrier boards via JLCPCB or PCBWay turnkey assembly the moment the schematic and bill-of-materials are released. Cost <£500 for three boards. This validates the layout and RF performance long before the body tooling decision needs to be final.
- **Specify IP54 at module level, IP65 at body level.** Write the seal spec into the module data sheet so Chase can source the right gaskets and the right contract manufacturer.
- **Plan three radio-compliance test campaigns:** base SKU (Wi-Fi/Bluetooth-Low-Energy + LoRa), premium SKU (with cellular), and the regulatory delta if the body shielding affects emissions. The pre-certified modules cover their own compliance, but the integrated product still needs Conformité Européenne Radio Equipment Directive emissions testing — budget £6–10k and 4 weeks at a notified body.
- **Dual-source the LoRa transceiver.** Semtech SX1262 has had supply wobbles. Ai-Thinker Ra-01SH is a drop-in module-level alternative. *Never single-source anything critical.*
- **ASSEMBLY: Sequence — middle.** Build order: Power Subsystem ' Compute & Sensor Core ' **Connectivity Module** (mates via Hirose DF40) ' Enclosure close-up. Connectivity goes in after compute is electrically verified, before the body is sealed.
- **ASSEMBLY: Fixtures & alignment.** Need (a) a board-to-board mating fixture that aligns the DF40 connector without zero-insertion-force damage, (b) U.FL pigtail routing jig with strain-relief positioning, (c) torque-controlled antenna pigtail seating tool — U.FL is famously easy to mis-seat.
- **ASSEMBLY: Critical assembly tolerances.** DF40 mating perpendicularity ±0.2 mm, antenna pigtail length tolerance ±5 mm (longer pigtails detune at high frequency), antenna aperture-to-radiator clearance ±1 mm.
- **ASSEMBLY: Test points.** Immediately after module assembly: (a) current draw at idle <50 mA, (b) Wi-Fi scan returns >3 networks, (c) LoRa transmit at +14 dBm verified on spectrum analyser, (d) cellular registration on test SIM (premium variant only), (e) U.FL connector continuity test before pigtail routing. Total in-circuit test time target: <60 seconds per board.
- **ASSEMBLY: Dependencies.** Compute & Sensor Core must be assembled and basic-functional before Connectivity mates. Antenna apertures in the body must be machined/molded and dielectric windows installed before final close-up. Power Subsystem must deliver clean 3.8 V rail before cellular transmit testing.

4. BOM master (70 rows)

BOM derived from the module decomposition's keyParts, expanded into typed part rows. Part records live in the `parts` table and are joined back to modules via source_module_id.

PART #	NAME	MODULE	TYPE	PROCESS / MATERIAL	MASS	COST
CMP-001	Main PCBA — Main 6-layer PCBA carrying the NPU, application processor, memory, sensor interfaces, and connectors for camera, microphone, and environmental modules. Populated and tested as the primary compute board of the feeder.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	make	other · FR4 epoxy glass laminate, mixed SMT components · ±±0.1mm outline, ±10% impedance	0.08kg	£65
CMP-002	Main PCB Bare Board — Main 6-layer FR-4 PCB carrying the NPU, camera connector, memory, microphone, and environmental sensor with controlled-impedance routing for high-speed MIPI CSI-2 and LPDDR4 buses.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	make	other · FR-4 Glass Epoxy · ±±0.1mm board outline, ±10% impedance	0.04kg	£38
CMP-003-PUR	NPU SoC (RV1106 / Hailo-8L) — Edge AI NPU SoC providing ~5 TOPS for on-device bird species and individual identification via vision models. Rockchip RV1106 with integrated 1 TOPS NPU plus optional Hailo-8L co-processor for higher throughput.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	buy	—	0.00kg	£18
CMP-004-PUR	CMOS Image Sensor (IMX577) — Sony IMX577 12.3 MP back-illuminated CMOS image sensor providing high dynamic range stills and 4K video for bird identification at the perch.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	buy	—	0.00kg	£32
CMP-005	Custom 3x Optical Zoom Lens Module — Custom 3x optical zoom lens module with stepper-driven zoom and focus groups, IR-cut filter, and weather-sealed barrel matched to the IMX577 sensor for capturing bird detail at 1-3 m.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	make	other · Aluminium barrel with multi-element optical glass · ±±0.02mm lens spacing, ±0.1mm mount	0.06kg	£75
CMP-006-PUR	LPDDR4 SDRAM — LPDDR4 SDRAM providing high-bandwidth working memory for the NPU and image processing pipeline. 2 GB capacity sufficient for buffering 4K frames and model weights.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	buy	—	0.00kg	£10
CMP-007-PUR	eMMC 5.1 Flash — eMMC 5.1 NAND flash storing OS, AI models, and rolling buffer of bird visit logs and highlight clips. 32 GB capacity supports multi-month local retention.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	buy	—	0.00kg	£7

CMP-008-PUR	Digital MEMS Microphone (SPH0641LU4H-1) — Digital MEMS microphone with PDM output and ultrasonic-capable bandwidth for bird vocalization capture and species classification audio assist.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	buy	—	0.00kg	£2
CMP-009-PUR	Environmental Sensor (BME280) — Bosch BME280 combined temperature, humidity, and barometric pressure sensor providing ambient environmental data correlated with bird visits and used for solar/charge thermal lockout logic.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	buy	—	0.00kg	£6
CMP-010-PUR	Lux Sensor (TSL2591) — Ambient light sensor with high dynamic range for visible and IR light measurement, used to gate camera exposure and detect day/night conditions for power management.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	buy	—	0.00kg	£6
CMP-011-PUR	Auxiliary MCU (STM32L4) — Low-power auxiliary microcontroller handling always-on sensor polling, perch load cell sampling, and wake-up signalling to the main NPU SoC during sleep.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	buy	—	0.00kg	£4
CMP-012-PUR	Load Cell ADC (HX711) — 24-bit ADC load cell amplifier (HX711) interfacing the perch strain-gauge bridge to the auxiliary MCU for high-resolution mass measurement.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	buy	—	0.00kg	£2
CMP-013	Aluminium Thermal Spreader — CNC-machined aluminium thermal spreader bonded to the NPU SoC and camera ISP to conduct heat into the enclosure wall, maintaining sensor performance in summer sun.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	make	cnc · 6061 Aluminium · ±±0.05mm	0.04kg	£15
CMP-ASY	Compute & Sensor Core Assembly — Top-level assembly representing the integrated compute and sensor core, comprising the main PCBA, NPU, camera, microphone, and environmental sensors. Acts as the brain of the feeder for on-device species and individual identification.	Compute & Sensor Core (NPU + Camera + Mic + Environmental)	make	other · Mixed (PCBA assembly) · ±±0.3mm assembly	0.12kg	£95
CON-001	Connectivity Carrier PCBA — Populated connectivity carrier PCB with Wi-Fi/BLE, cellular, and LoRa radio modules, level shifters, SIM holder, and U.FL antenna connectors. Interfaces to the compute core via board-to-board connector.	Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)	make	other · FR-4 PCB with SMT components · ±±0.1mm component placement	0.04kg	£38

CON-002	Connectivity Carrier Bare PCB — Bare 4-layer carrier PCB providing routing, RF controlled-impedance traces, and ground pour for the connectivity module. Includes antenna keepouts and module footprints for ESP32-C6, BG95-M3, and SX1262.	Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)	make	other · FR-4 · ±±0.1mm	0.02kg	£7
CON-003-PUR	Wi-Fi/BLE Module (ESP32-C6-WROOM-1) — Espressif ESP32-C6 Wi-Fi 6 and Bluetooth LE 5.3 module providing primary connectivity to the home network. Pre-certified RF module with PCB antenna, soldered as SMT component to the carrier.	Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)	buy	—	0.00kg	£3
CON-004-PUR	Cellular Module (Quectel BG95-M3) — Quectel BG95-M3 multi-mode LTE Cat M1/NB2/EGPRS cellular module providing fallback connectivity where Wi-Fi is unavailable. LGA package soldered to the carrier with SIM and U.FL antenna interface.	Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)	buy	—	0.00kg	£19
CON-005-PUR	LoRa Transceiver (SX1262) — Semtech SX1262 sub-GHz LoRa transceiver IC for long-range mesh networking between feeders on 868MHz EU band. QFN package with external matching network and U.FL antenna connector on the carrier.	Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)	buy	—	0.00kg	£6
CON-006-PUR	Wi-Fi PCB Antenna (W3006) — Surface-mount 2.4/5 GHz Wi-Fi PCB chip antenna for the connectivity module, providing dual-band coverage to home networks. Mounted near the dielectric aperture in the enclosure for unobstructed RF emission.	Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)	buy	—	0.00kg	£1
CON-007-PUR	LTE Flex Antenna (TG.30.8113) — Flexible PCB LTE/4G antenna (Taoglas TG.30.8113) covering 698-2700 MHz cellular bands for fallback connectivity. Adhesive-mounted to interior of dielectric aperture with 100mm IPEX cable to module.	Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)	buy	—	0.00kg	£7
CON-008-PUR	LoRa Sub-GHz Antenna — 868 MHz Sub-GHz dipole antenna for LoRa mesh communication between feeders. SMA-connected with right-angle mount through dielectric aperture in enclosure.	Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)	buy	—	0.01kg	£4
CON-009-PUR	Sub-GHz RF Front-End (SKY66423-11) — Skyworks SKY66423-11 Sub-GHz RF front-end module with integrated PA, LNA, and T/R switch for 868/915 MHz LoRa transmission. Provides +30dBm output power for extended mesh range.	Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)	buy	—	0.00kg	£4

CON-010-PUR	Board-to-Board Connector (Hirose DF40) — Hirose DF40 series 0.4mm pitch board-to-board connector pair linking the connectivity module to the compute core PCB. Provides high-density signal and power interconnect with 1.5mm stack height.	Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)	buy	—	0.00kg	£2
CON-ASY	Connectivity Module Assembly — Top-level assembly of the connectivity module bringing together the carrier PCBA, radio modules, antennas, and shielding. Provides Wi-Fi/BLE, optional cellular, and LoRa mesh radios to the feeder.	Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)	make	other · Mixed Assembly · ±±0.5mm assembly stack	0.09kg	£48
ENC-001	Powder-Coated Steel Outer Shell — Powder-coated steel outer shell forming the primary weatherproof body of the feeder. Provides UV/water resistance and impact protection with mounting bosses for the solar roof, perch, and hanging bracket.	Primary Enclosure & Mounting Structure	make	sheet_metal · Cold-rolled mild steel · ±±0.3mm	1.10kg	£28
ENC-002	Stoneware Ceramic Body (Premium Variant) — Premium variant stoneware ceramic body, slip-cast and high-fired for a frost-proof, garden-friendly aesthetic finish. Replaces the steel shell on premium SKUs while maintaining identical mounting interfaces.	Primary Enclosure & Mounting Structure	make	casting · Stoneware ceramic · ±±1.5mm (post-fire shrinkage)	1.80kg	£42
ENC-003	Galvanised Steel Inner Sub-Chassis — Galvanised steel inner sub-chassis that carries the PCB stack, battery tray and reservoir dock interface. Mechanically decouples electronics from the outer shell and provides EMC reference plane.	Primary Enclosure & Mounting Structure	make	sheet_metal · Hot-dip galvanised steel · ±±0.2mm	0.55kg	£15
ENC-004	Dielectric Antenna Window Insert — RF-transparent dielectric window insert moulded into the upper rear of the shell to allow Wi-Fi/BLE/LoRa antenna radiation. Sealed to the shell with a co-moulded EPDM lip and UV-stabilised for outdoor use.	Primary Enclosure & Mounting Structure	make	injection_molding · ASA polymer · ±±0.15mm	0.04kg	£2
ENC-005-PUR	Gore Acoustic Vent GAW304 — Gore protective acoustic vent GAW304 providing IP67 pressure equalisation and water/dust ingress protection for the microphone and enclosure interior. Adhesive-mounted over a vent hole in the shell.	Primary Enclosure & Mounting Structure	buy	—	0.00kg	£4
ENC-006	Stainless Acoustic Port Mesh — Fine stainless steel mesh disc protecting the microphone acoustic port from insects and debris while remaining acoustically transparent. Press-fit into a recess behind the Gore vent.	Primary Enclosure & Mounting Structure	make	sheet_metal · 304 Stainless Steel · ±±0.1mm	0.00kg	£0

ENC-007	EPDM Perimeter Gasket — Compression-moulded EPDM perimeter gasket sealing the outer shell to the lower service hatch / base plate. Provides IP54 weather sealing across UK temperature range.	Primary Enclosure & Mounting Structure	make	other · EPDM rubber · ±±0.3mm	0.06kg	£3
ENC-008	Hanging Bracket with Pole-Mount Adapter — Sheet-metal hanging bracket with integrated pole-mount adapter that allows the feeder to be suspended from a hook or clamped to a 25-50mm garden pole. Bolts to the rear of the primary enclosure via PEM nut inserts.	Primary Enclosure & Mounting Structure	make	sheet_metal · Stainless Steel · ±±0.3mm formed, ±0.1mm holes	0.42kg	£15
ENC-ASY	Primary Enclosure Assembly — Top-level primary enclosure assembly comprising outer shell, inner chassis, antenna window, gaskets, vents and acoustic mesh. Provides weatherproof IP54 housing with mechanical interfaces for reservoirs, perch, solar roof and hanging mount.	Primary Enclosure & Mounting Structure	make	other · Mixed assembly · ±±0.5mm	2.40kg	£95
FAS-001-PUR	A4 Stainless M4 Socket-Cap Fastener — A4 (316) stainless steel M4 socket-cap head screw used for primary enclosure and bracket fastening. Marine-grade corrosion resistance suitable for outdoor UK garden exposure.	Primary Enclosure & Mounting Structure	buy	—	0.00kg	£0
FAS-002-PUR	A4 Stainless M3 Fastener — A4 (316) stainless steel M3 socket-cap fastener used for securing internal sub-assemblies, PCB mounts, and gasket retainers. Marine-grade corrosion resistance for long outdoor service life.	Primary Enclosure & Mounting Structure	buy	—	0.00kg	£0
FAS-003-PUR	PEM M3/M4 Nut Insert — Self-clinching PEM nut insert pressed into the sheet-metal enclosure to provide reusable threaded fastening points for M3 and M4 screws. Compatible with 1.5-2.5mm stainless sheet.	Primary Enclosure & Mounting Structure	buy	—	0.00kg	£0
HEDGEROW-ASY	Hedgerow Top-Level Product Assembly — Top-level Hedgerow smart bird feeder product assembly integrating enclosure, solar roof, power subsystem, compute/sensor core, instrumented perch, modular reservoirs, and connectivity module. Final tested unit shipped to UK garden customers.	Primary Enclosure & Mounting Structure	make	other · Mixed assembly (ceramic, steel, PCB, LiFePO4) · ±±1.0mm assembly	4.20kg	£320
PER-001-PUR	Parallelogram Strain-Gauge Load Cell — Parallelogram aluminium strain-gauge load cell rated for small-bird mass measurement, providing repeatable mV/V output to the HX711 amplifier with low cross-axis sensitivity.	Instrumented Perch with Load Cell	buy	—	0.03kg	£7

PER-002	Stainless Steel Perch Bar — Machined stainless steel perch bar that birds land on, sized for typical UK garden species grip and corrosion-resistant for outdoor use, mounted to the free end of the load cell.	Instrumented Perch with Load Cell	make	machining · 304 Stainless Steel · ±±0.1mm	0.11kg	£8
PER-003	Silicone Perch Isolation Gasket — Compression-moulded silicone gasket isolating the perch load cell mounting from the main enclosure to reduce vibration cross-coupling and seal against water ingress.	Instrumented Perch with Load Cell	make	other · Silicone Rubber · ±±0.2mm	0.01kg	£1
PER-004-PUR	Shielded Twisted-Pair Signal Cable — Shielded twisted-pair cable carrying load cell signals from the perch strain gauge to the compute core, providing immunity to EMI from radios and switching converters.	Instrumented Perch with Load Cell	buy	—	0.03kg	£2
PER-005-PUR	Cable Grommet — Elastomeric cable grommet that seals the cable pass-through between the perch and primary enclosure, preventing water and debris ingress while providing strain relief.	Instrumented Perch with Load Cell	buy	—	0.00kg	£0
PER-ASY	Instrumented Perch Assembly — Top-level assembly for the instrumented perch sub-module, integrating the load cell, perch bar, isolation gasket, and fasteners into a serviceable unit that bolts to the main enclosure.	Instrumented Perch with Load Cell	make	other · Mixed assembly · ±±0.3mm assembly	0.18kg	£28
PWR-001	Power Subsystem PCBA — Power subsystem PCBA carrying the BMS AFE, buck regulator, MPPT input stage, current sensing, and protection circuitry. Routes solar input to battery and regulated rails to downstream modules.	Power Subsystem (LiFePO4 Battery + PMIC)	make	other · FR-4 PCB with SMT components · ±IPC-A-610 Class 2	0.04kg	£22
PWR-002-PUR	LiFePO4 Prismatic Cell — Serviceable LiFePO4 prismatic cell providing safe, long-cycle-life energy storage suitable for UK outdoor temperature range. Chosen for thermal stability and tolerance to partial state-of-charge cycling.	Power Subsystem (LiFePO4 Battery + PMIC)	buy	—	0.18kg	£11
PWR-003-PUR	BMS Analog Front-End IC (BQ76920) — Texas Instruments BQ76920 battery monitor and protection AFE for 3-5S Li-ion/LiFePO4 packs. Provides cell voltage monitoring, coulomb counting, and over/under-voltage and temperature protection via I2C.	Power Subsystem (LiFePO4 Battery + PMIC)	buy	—	0.00kg	£3

PWR-004-PUR	Buck Regulator IC (TPS62933) — Texas Instruments TPS62933 3.8-30V, 3A synchronous buck converter regulating battery voltage to a 5V or 3.3V system rail. High efficiency at light loads suits intermittent NPU and radio activity.	Power Subsystem (LiFePO4 Battery + PMIC)	buy	—	0.00kg	£1
PWR-005-PUR	Buck Regulator IC (TPS54360) — Synchronous step-down buck regulator IC providing efficient conversion from battery/solar input to 5V and 3.3V rails. Selected for wide input range and low quiescent current suitable for UK winter solar harvesting.	Power Subsystem (LiFePO4 Battery + PMIC)	buy	—	0.00kg	£3
PWR-006-PUR	NTC Thermistor — NTC thermistor used by the PMIC to monitor LiFePO4 cell pack temperature and enforce cold-charge lockout below 0°C. Mounted in thermal contact with the battery cradle.	Power Subsystem (LiFePO4 Battery + PMIC)	buy	—	0.00kg	£0
PWR-007-PUR	Polyfuse and TVS Input Protection Set — Combined polyfuse and TVS diode set protecting the power input from overcurrent and surge events from the solar panel and external charge path. Provides resettable overcurrent and clamping for ESD/lightning-induced transients.	Power Subsystem (LiFePO4 Battery + PMIC)	buy	—	0.00kg	£1
PWR-008	Battery Cradle — Injection-molded battery cradle that locates and retains the serviceable LiFePO4 cell pack within the enclosure. Features integrated guide ribs, contact mounts, and a thermistor pocket.	Power Subsystem (LiFePO4 Battery + PMIC)	make	injection_molding · Glass-filled Polyamide · ±±0.2mm	0.06kg	£2
PWR-009	Cradle Sprung Contact Set — Sprung sheet-metal contact set that connects the LiFePO4 cells to the power PCB via the cradle. Provides reliable, vibration-tolerant contact pressure across cell tolerance stack-up.	Power Subsystem (LiFePO4 Battery + PMIC)	make	sheet_metal · Phosphor Bronze · ±±0.1mm	0.01kg	£1
PWR-010	Power PCB Bare Board — Bare 4-layer FR4 PCB for the power subsystem hosting the buck regulator, protection circuitry, and battery management. Heavy copper for current-carrying traces and conformal-coating compatible.	Power Subsystem (LiFePO4 Battery + PMIC)	make	other · FR4 epoxy glass laminate · ±±0.1mm outline, ±10% impedance	0.03kg	£7
PWR-ASY	Power Subsystem Assembly — Complete power subsystem assembly integrating the LiFePO4 battery pack, PMIC PCBA, BMS, and wiring harness. Delivers regulated rails to all electronic modules and manages charge from the solar roof with cold-charge lockout.	Power Subsystem (LiFePO4 Battery + PMIC)	make	other · Mixed (PCBA + LiFePO4 cells + harness) · ±±0.5mm	0.45kg	£65

RES-001	Seed Reservoir Body (Tritan) — Transparent injection-moulded reservoir body that holds seed and gravity-feeds it to the feed port, allowing visual seed level inspection and bayonet engagement to the dock.	Modular Reservoir Dock & Seed Reservoirs	make	injection_molding · Tritan Copolyester · ±±0.2mm	0.18kg	£7
RES-002	Bayonet Dock Lug Set — Set of three machined bayonet lugs that engage the reservoir body to the dock with a quarter-turn locking action, providing mechanical retention against gravity and vibration.	Modular Reservoir Dock & Seed Reservoirs	make	machining · 316 Stainless Steel · ±±0.05mm	0.03kg	£5
RES-003-PUR	Hall-Effect Sensor (A1126) — Hall-effect sensor that detects the reservoir ID magnet key to identify which seed type (nyjer/sunflower/suet) is installed, signalling the compute core via digital output.	Modular Reservoir Dock & Seed Reservoirs	buy	—	0.00kg	£2
RES-004	Reservoir ID Magnet Key — Encapsulated neodymium magnet pressed into each reservoir body at a coded angular position, providing reservoir-type identification to the Hall sensor in the dock.	Modular Reservoir Dock & Seed Reservoirs	make	other · NdFeB Neodymium Magnet · ±±0.1mm	0.00kg	£1
RES-005	Stainless Feed Port Grille — Stainless steel sheet metal grille at the feed port that meters seed flow, prevents large debris ingress, and resists pecking damage from birds.	Modular Reservoir Dock & Seed Reservoirs	make	sheet_metal · 304 Stainless Steel · ±±0.2mm	0.03kg	£3
RES-006-PUR	Anti-Squirrel Weighted Closure Spring — Weighted compression spring used in the reservoir dock closure mechanism to resist squirrel tampering by requiring sustained force to displace. Returns the closure to sealed position when load is removed.	Modular Reservoir Dock & Seed Reservoirs	buy	—	0.03kg	£2
RES-007	Reservoir Dock Seal — Compression gasket seal between reservoir base and dock interface preventing moisture ingress to seed and feed mechanism. Custom-cut profile matches keyed dock geometry.	Modular Reservoir Dock & Seed Reservoirs	make	other · Silicone Rubber · ±±0.3mm	0.01kg	£2
RES-ASY	Reservoir Dock & Reservoirs Assembly — Top-level assembly of the reservoir dock, including dock body, bayonet lugs, Hall sensor, ID magnet, and feed grille, providing keyed quick-release of interchangeable seed reservoirs.	Modular Reservoir Dock & Seed Reservoirs	make	other · Mixed (assembly) · ±±0.5mm assembly	0.42kg	£22
SOL-001	Custom Monocrystalline PV Laminate — Custom monocrystalline silicon PV laminate sized for the feeder roof, delivering ~10W peak under UK winter low-irradiance conditions. ETFE front sheet with EVA encapsulant and back-sheet for outdoor durability.	Solar Roof Assembly	make	other · Monocrystalline Silicon PV Laminate · ±±1.0mm trim, ±5% Pmax	0.55kg	£32

SOL-002	Tilt Frame — Cast aluminium tilt frame that holds the PV laminate at a fixed UK-optimised tilt angle (~50°) and provides the bolted interface to the enclosure top. Integral ribs stiffen the perimeter and form the gasket channel.	Solar Roof Assembly	make	casting · Aluminium Alloy · ±±0.3mm machined faces, ±0.5mm cast	0.70kg	£18
SOL-003	Solar Roof Perimeter Gasket — Closed-cell EPDM perimeter gasket that seals the PV laminate to the tilt frame and prevents water ingress to the enclosure. Die-cut continuous loop with self-adhesive backing.	Solar Roof Assembly	make	other · EPDM Rubber · ±±0.5mm profile	0.04kg	£3
SOL-004-PUR	IP67 DC Weatherproof Connector — IP67-rated weatherproof DC connector pair for routing solar panel output through the enclosure wall to the power subsystem. Provides sealed, serviceable interconnect rated for outdoor UV and moisture exposure.	Solar Roof Assembly	buy	—	0.03kg	£3
SOL-005-PUR	Schottky Blocking Diode — Schottky blocking diode preventing reverse current flow from the battery into the solar panel during low-light or night conditions. Low forward voltage drop minimizes harvesting losses.	Solar Roof Assembly	buy	—	0.00kg	£0
SOL-006-PUR	Bonded Sealing Washer Set — Set of bonded sealing washers (rubber-to-metal) used under fasteners passing through the solar roof and enclosure to prevent water ingress. Ensures IP67 integrity at mechanical penetrations.	Solar Roof Assembly	buy	—	0.00kg	£0
SOL-ASY	Solar Roof Assembly — Top-level solar roof assembly comprising the PV laminate, cast tilt frame, perimeter gasket, and fasteners. Bolts to the top of the primary enclosure and delivers DC output to the power subsystem via a sealed gland.	Solar Roof Assembly	make	other · Assembly · ±±0.5mm assembly	1.85kg	£78

5. Cost waterfall

Cost estimates grounded against the material-properties library (47 rows, freshest 23 Mar 2026) and the `process_capabilities` table (20 rows, freshest unknown). Costs sit in the project's ai_cost_estimates jsonb, keyed by module id.

UNIT COST (ALL-IN)	CEILING (BRIEF)	HEADROOM
£354	**£155**	**£-199**

Per-module roll-up

MODULE	COST	% OF UNIT
Primary Enclosure & Mounting Structure	£60	16.8%
Solar Roof Assembly	£38	10.8%
Power Subsystem (LiFePO4 Battery + PMIC)	£57	16.2%
Compute & Sensor Core (NPU + Camera + Mic + Environmental)	£107	30.1%
Instrumented Perch with Load Cell	£19	5.4%
Modular Reservoir Dock & Seed Reservoirs	£24	6.9%
Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)	£49	13.9%

6. Risks register

Every failure mode and open question declared against each module, in one register. Until a dedicated risks store ships with severity + ownership, these rows are the canonical risk inventory.

Primary Enclosure & Mounting Structure

KNOWN FAILURE MODES (4)

- Powder-coat chip-through at edges leading to substrate corrosion and rust bleed within 2–3 UK winters
- Ceramic body cracking from freeze-thaw water ingress at unglazed mounting points or differential thermal expansion against the steel sub-chassis
- Gasket compression set after thermal cycling causing IP54 breach and water ingress to the compute PCB
- Antenna window adhesive (VHB) creep in summer roof temperatures (>60 °C) leading to dielectric insert detachment

OPEN QUESTIONS (3)

- Final body geometry and overall W×D×H envelope — driven by reservoir volume, battery format, and camera standoff, all pending CAD
- Whether to commit to ceramic for the premium tier or use a ceramic-look glaze on steel (lower tooling risk, lower mass, simpler RF)
- Confirmed IP rating target — IP54 sufficient for sheltered garden use, or IP65 required for exposed installations and warranty claims

Solar Roof Assembly

KNOWN FAILURE MODES (4)

- Seal degradation at gasket interface allowing water ingress into the enclosure, corroding electronics and shorting the PV output
- ETFE/encapsulant delamination after 3-5 years UV exposure, reducing output and causing visible yellowing that damages the premium aesthetic
- Soiling and bird-droppings on panel surface reducing winter yield by 20-40% below the energy-balance threshold
- Micro-cracking of silicon cells from thermal cycling (-15 °C to +45 °C) or hail impact, causing hot-spot failures

OPEN QUESTIONS (3)

- Final panel area and Wp rating pending energy-balance simulation across UK Met Office irradiance data for worst-case 52°N December conditions
- Whether to specify a single fixed tilt or offer two SKUs (north-facing garden vs south-facing garden) with different tilt frames
- Self-cleaning strategy: hydrophobic coating vs steeper tilt vs user-maintenance prompt in app

Power Subsystem (LiFePO4 Battery + PMIC)

KNOWN FAILURE MODES (4)

- Lithium plating and permanent capacity loss if charge-temperature interlock fails and cells charge below 0 °C during UK winter
- Deep-discharge cell damage during prolonged overcast periods if low-voltage cutoff or sleep-mode quiescent current budget is mis-sized
- Moisture ingress to PCB causing dendrite growth across high-impedance BMS sense lines, leading to false cell-imbalance shutdowns
- Connector corrosion at battery cradle contacts after 3–5 years of outdoor thermal cycling, causing intermittent power loss

OPEN QUESTIONS (3)

- Final battery capacity (Wh) — requires energy-budget model of NPU duty cycle vs. December UK solar yield at site latitude
- Whether a self-heating circuit is needed for charging in sub-zero conditions, or whether lockout-and-wait is acceptable for the autonomy target
- Cellular-variant power budget — LTE-M TX bursts may require larger bulk capacitance and a higher-current 3.8 V rail not yet specified

Compute & Sensor Core (NPU + Camera + Mic + Environmental)

KNOWN FAILURE MODES (4)

- NPU thermal throttling or shutdown under sustained inference inside ceramic body with poor convective cooling
- Lens fogging or condensation behind the front window in damp UK conditions, degrading image quality and ML accuracy
- Voice-coil-motor zoom mechanism wear or stiction after thousands of cycles in cold/wet outdoor service
- MIPI CSI-2 EMI coupling into Wi-Fi/LoRa antennas via the steel chassis, breaching CE-RED radiated emissions

OPEN QUESTIONS (3)

- Final NPU SoC selection (Hailo-8L vs Rockchip RV1106 vs Ambarella CV25) pending model-accuracy benchmarking on UK garden-bird dataset
- Required optical zoom range and minimum focus distance — depends on final camera-to-perch standoff set by enclosure CAD
- Whether passive conduction to body is sufficient or an internal heatsink/heat-pipe is needed at peak inference duty cycle

Instrumented Perch with Load Cell

KNOWN FAILURE MODES (4)

- Load cell zero drift with temperature across UK -15 °C to +45 °C range causing false individual reassignment; requires periodic auto-tare and temperature compensation
- Water ingress through perch-body interface corroding strain gauges and causing bridge imbalance or open circuit
- Mechanical overload from a perched corvid, squirrel, or user handling exceeding load cell rated capacity, causing permanent calibration shift
- Wind-induced perch oscillation coupling into the load cell signal and saturating the ADC during gusts, requiring low-pass filtering and gust rejection in firmware

OPEN QUESTIONS (3)

- Final load cell capacity selection — 100 g optimises resolution for small passerines but may be overloaded by woodpigeons or grey squirrels; 500 g cell sacrifices ~5x resolution
- Whether a single shared perch or one perch per reservoir is required for unambiguous mass-to-feed-port association
- Isolation gasket stiffness and gap geometry needed to maintain <1% crosstalk from body-borne loads (hanging sway, reservoir refill)

Modular Reservoir Dock & Seed Reservoirs

KNOWN FAILURE MODES (4)

- Seed bridging / clogging at feed port causing perceived empty reservoir while seed remains — especially nyjer in damp UK conditions
- Gasket UV degradation after 2–3 summers leading to rain ingress, seed germination inside reservoir, and mould
- Bayonet lugs wearing or fouling with seed dust causing accidental release during high winds, dropping reservoir and seed onto lawn
- Hall sensor false-negative if magnets demagnetise or shift, causing reservoir to read as 'absent' and disabling species priors

OPEN QUESTIONS (3)

- Final reservoir volume vs. refill-frequency UX trade-off (weekly refill target sets minimum capacity)
- Whether suet-ball cage variant needs its own dedicated load path or can share the standard dock with an adapter
- Dishwasher-cycle durability of printed/embossed species labelling on reservoir body

Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)

KNOWN FAILURE MODES (4)

- RF performance degradation due to steel body shielding if antenna apertures are mis-located or sealing gaskets compress the dielectric window
- Water ingress through U.FL pigtail routing or antenna apertures causing corrosion of RF connectors and detuning
- Cellular module thermal shutdown or brown-out during transmit bursts (peak ~2 A on LTE-M) if power subsystem decoupling is inadequate
- LoRa mesh range collapse in dense suburban RF environments or due to TCXO frequency drift across UK temperature extremes (-15 to +45 °C)

OPEN QUESTIONS (3)

- Final antenna aperture geometry and dielectric material — depends on body material decision (ceramic vs powder-coated steel)
- Whether cellular module is a factory-fit SKU split or a user-installable upgrade slot (affects connector cost and seal integrity)
- LoRa duty-cycle budget and mesh routing protocol (LoRaWAN private gateway vs proprietary mesh) for multi-feeder deployments

7. Supplier shortlist (21)

Shortlist built by scoring each supplier in the directory (651 companies) and marketplace listings (28,315 listings) against each module's declared process + material. A low shortlist count usually means the directory doesn't yet have coverage for the project's niche — not that no match exists globally.

SheetMetalPro Ltd

SheetMetalPro Ltd

HQ not declared · ramp role unassigned · match score 24.6

Matched across 1 module: Primary Enclosure & Mounting Structure

SCORE BREAKDOWN

total	24.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	21.2
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Manufacturer

pumpkinspace.com

MBM2 high-performance flight computer for CubeSats and nanosatellites.

https://www.pumpkinspace.com/store/p208/mbm2.html

Contact:

HQ not declared · ramp role unassigned · match score 23.4

Matched across 1 module: Compute & Sensor Core (NPU + Camera + Mic + Environmental)

SCORE BREAKDOWN

total	23.4
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

ceramic-substrates.co.uk

Ceramic Substrates

http://www.ceramic-substrates.co.uk/

Contact: info@ceramic-substrates.co.uk

HQ not declared · ramp role unassigned · match score 24.6

Matched across 1 module: Primary Enclosure & Mounting Structure

SCORE BREAKDOWN

total	24.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	21.2
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Products

mepcomouldings.co.uk

Mepco Mouldings LTD

https://mepcomouldings.co.uk/

Contact: enquires@mepcomouldings.co.uk

HQ not declared · ramp role unassigned · match score 24.7

Matched across 1 module: Primary Enclosure & Mounting Structure

SCORE BREAKDOWN

total	24.7
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	21.3
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Products

solfit.co.uk

Solfit

https://solfit.co.uk/

Contact: enquiries@solfit.co.uk

HQ not declared · ramp role unassigned · match score 28.3

Matched across 1 module: Solar Roof Assembly

SCORE BREAKDOWN

total	28.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	27.9
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Products

gtc.co.uk

RockBLOCK Pro OEM – Satellite IoT Gateway for Custom Integration

https://gtc.co.uk/products/rockblock-oem-pro

Contact: info@gtc.co.uk

HQ not declared · ramp role unassigned · match score 23.8

Matched across 1 module: Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)

SCORE BREAKDOWN

total	23.8
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.4
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

sunfixings.co.uk

Sunfixings Ltd

https://www.sunfixings.co.uk

Contact: sales@sunfixings.co.uk

HQ not declared · ramp role unassigned · match score 27.0

Matched across 1 module: Solar Roof Assembly

SCORE BREAKDOWN

total	27
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.6
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturer

cmxbattery.com

LiFePo4 19" Rack-Mount Li-Ion Lithium iron phosphate server rack Battery

https://cmxbattery.com/Product/19-rack-mount-li-ion-battery-with-lithium-iron-phosphate-technology/

Contact: sales@cmxbattery.com

HQ not declared · ramp role unassigned · match score 31.5

Matched across 1 module: Power Subsystem (LiFePO4 Battery + PMIC)

SCORE BREAKDOWN

total	31.5
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	31.1
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

coremax-tech.com

lifepower4 48v sever rack mount solar lithium lifepo4 battery bank

https://coremax-tech.com/product/48v-100ah-lifepo4-server-rack-battery-storage/

Contact: info@coremax-tech.com

HQ not declared · ramp role unassigned · match score 29.0

Matched across 1 module: Power Subsystem (LiFePO4 Battery + PMIC)

SCORE BREAKDOWN

total	29
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	28.6
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

fischerconnectors.com

Fischer Connectors

https://fischerconnectors.com/en/

Contact: reach@fischerconnectors.ch

HQ not declared · ramp role unassigned · match score 20.9

Matched across 1 module: Modular Reservoir Dock & Seed Reservoirs

SCORE BREAKDOWN

total	20.9
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	17.5
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Products

maxwellenergy.co

Battery Management System & BMS for High Power Systems

https://www.maxwellenergy.co/blog/battery-management-system-bms-for-high-power-systems-the-complete-guide/

HQ not declared · ramp role unassigned · match score 29.0

Matched across 1 module: Power Subsystem (LiFePO4 Battery + PMIC)

SCORE BREAKDOWN

total	29
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	28.6
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

xmos.com

XMOS

https://www.xmos.com

Contact: sales@xmos.com

HQ not declared · ramp role unassigned · match score 24.2

Matched across 1 module: Compute & Sensor Core (NPU + Camera + Mic + Environmental)

SCORE BREAKDOWN

total	24.2
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.8
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Hardware Company

ems-uk.com

ems-uk

https://ems-uk.com/

Contact: sales@ems-uk.com

HQ not declared · ramp role unassigned · match score 25.0

Matched across 1 module: Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)

SCORE BREAKDOWN

total	25
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	21.6
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Manufacturing

avgauge.com

AV Gauge & Fixture Inc.

http://www.avgauge.com/index

Contact: info@avgauge.com

HQ not declared · ramp role unassigned · match score 20.6

Matched across 1 module: Instrumented Perch with Load Cell

SCORE BREAKDOWN

total	20.6
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	17.2
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Gauging & Check Fixtures

airborne.ed.ac.uk

PAR Sensors

https://airborne.ed.ac.uk/airborne-research-and-innovation/airborne-sensors/meteorogical-systems/par-sensors

HQ not declared · ramp role unassigned · match score 20.4

Matched across 1 module: Instrumented Perch with Load Cell

SCORE BREAKDOWN

total	20.4
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	20
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Manufacturing

weptech.de

WEPTECH elektronik GmbH

https://www.weptech.de/en/ems.html

Contact: sales@weptech.de

HQ not declared · ramp role unassigned · match score 23.7

Matched across 1 module: Connectivity Module (Wi-Fi/BLE + optional Cellular + LoRa)

SCORE BREAKDOWN

total	23.7
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	23.3
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Electronics Manufacturing Services (EMS)

leadmetrology.com

LEAD METROLOGY

https://www.leadmetrology.com/

Contact: sales@leadmetrology.com

HQ not declared · ramp role unassigned · match score 20.2

Matched across 1 module: Instrumented Perch with Load Cell

SCORE BREAKDOWN

total	20.2
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	16.8
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Manufacturing

essentracomponents.com

Essentra Components

https://www.essentracomponents.com/en-gb

Contact: sales@essentracomponents.com

HQ not declared · ramp role unassigned · match score 20.9

Matched across 1 module: Modular Reservoir Dock & Seed Reservoirs

SCORE BREAKDOWN

total	20.9
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	17.5
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Fasteners and Cable Management

heatable.co.uk

Heatable

https://heatable.co.uk/solar

Contact: sales@heatable.co.uk

HQ not declared · ramp role unassigned · match score 28.8

Matched across 1 module: Solar Roof Assembly

SCORE BREAKDOWN

total	28.8
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	28.4
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Services

raptorphotonics.com

Raptor Photonics Ltd

https://www.raptorphotonics.com

Contact: sales@raptorphotonics.com

HQ not declared · ramp role unassigned · match score 23.3

Matched across 1 module: Compute & Sensor Core (NPU + Camera + Mic + Environmental)

SCORE BREAKDOWN

total	23.3
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	19.9
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Manufacturer

selectronix.co.uk

Selectronix Onboard Ltd

https://selectronix.co.uk/

Contact: sales@selectronix.co.uk

HQ not declared · ramp role unassigned · match score 22.1

Matched across 1 module: Modular Reservoir Dock & Seed Reservoirs

SCORE BREAKDOWN

total	22.1
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	18.7
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Manufacturing